Dear Editorial Team,
Communications AI & Computing

We are pleased to submit our manuscript, *"***Reviving, reproducing, and revisiting Axelrod's second tournament***,"* by Vincent Knight, Owen Campbell, Marc Harper, T. J. Gaffney, and Nikoleta E. Glynatsi, for consideration in *Communications AI & Computing*.

Robert Axelrod's Iterated Prisoner's Dilemma (IPD) tournaments, run in the early 1980s, are landmark studies that helped establish the modern field of the evolution of cooperation. These tournaments showed how simple reciprocal strategies can outperform more complex alternatives and have motivated a large body of subsequent work in game theory, evolutionary biology, economics, and political science. Today, hundreds of papers each year still cite and build on Axelrod's conclusions.

Although groundbreaking at the time, Axelrod's analysis reflects the computational and methodological constraints of the early 1980s. Since then, the field has changed substantially:

- there have been dramatic increases in computational power,
- new insights into repeated-game dynamics,
- the development of libraries with rich pools of strategies,
- and modern standards for transparency, reproducibility, and open science.

These developments make it timely to re-examine key conclusions from Axelrod's tournaments using a broader strategic landscape and contemporary methodological tools.

In this study, we analyze and extend Axelrod's famous IPD tournaments to reassess long-standing conclusions about strategic success, especially the dominance of Tit For Tat. By placing the original strategies into a much richer strategic landscape, we conduct the largest IPD tournament reported to date (272 strategies) and analyse millions of systematic variants of Axelrod's setup.

This work provides the first complete and transparent reconstruction of Axelrod's second tournament. We recover and modernise the original Fortran strategies, develop a Python interface that allows them to run unmodified within a widely used open-source research framework, and systematically reproduce Axelrod's published methodology.

We believe this manuscript is well suited for *Nature Communications* because it (i) revisits a landmark study on the evolution of cooperation using modern tools and data, (ii) confirms the original findings while showing that they do not fully hold in richer strategic settings, and (iii) tackles a longstanding reproducibility problem in computational social science by recovering and modernising key software artefacts. More broadly, our results refine a foundational understanding of cooperation in repeated games, a topic that remains important across both the natural and social sciences.

As potential referees, we would like to suggest the following:

- **Julian García** (julian.garcia@monash.edu, Monash University) is a computer scientist who is an expert in computational social science as well as in evolutionary game dynamics and the study of cooperation.
- **Jiawei Li** (Jiawei.Li@nottingham.edu.cn, University of Nottingham Ningbo China) works on computational approaches to strategic behaviour and repeated games.
- **Seung Ki Baek** (seungki.baek@gmail.com, Pukyong National University) has published extensively on evolutionary game theory and models of reciprocity.
- **Xingru Chen** (xingrucz@gmail.com, Beihang University) is a mathematician with a strong interest in direct reciprocity.
- **Ted Turocy (**T.Turocy@uea.ac.uk, University of East Anglia) is the theme lead at The Alan Turing Institute for the project Automated analysis of strategic interactions which has a goal of enhancing the reproducibility of the theoretical and empirical analysis of games.

Thank you for considering our submission.

With kind regards,
On behalf of the authors,

Professor Vincent Knight